

The Hand That Gives

*Pre-PIE *de- as Arm-Extension: the Directional Particle, the Body-Part
Heteroclite, and the Give-Verb *deh₃-*

Paper 1 of the *der-* project
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A Note on Notation

*This paper uses the traditional reconstruction *de- for the Pre-PIE hand-element throughout. Paper 3 of the project argues that a Pre-PIE unity may underlie surface PIE *de- and surface PIE *d^her- “hold, support, carry” (Sanskrit dharma, Latin firmus, Greek θεραπ-) and *d^herb^h- (Lithuanian darbas, dirbti; Sanskrit darbha-). At the PIE level these remain separate roots; the deeper unity is not load-bearing here.*

ABSTRACT

This paper develops three derivational outputs of a Pre-PIE arm-extension element *de- in the post-Anatolian Indo-European tradition. The first is the directional/allative particle *de/do, preserved most clearly in Greek -δε, productive in Mycenaean Linear B by 1400–1200 BCE. The second is the body-part heteroclite arm-noun *de-r-, whose daughter-language reflexes — Albanian *dorë*, Old Irish *derna*, Greek δάρης — display the ARM → HAND grammaticalization attested within Sanskrit and articulated for Proto-Celtic by Matasóvič (2009); a sister s-stem heteroclite *dóws- generates the Sanskrit *doṣ-/dor-*, Avestan *daoš-*, and Balto-Slavic *pa-duse/pazuxa* arm-vocabulary. The third is the post-Anatolian give-verb *deh₃-, analyzed bimorphemically as *de- + *-h₃-, with the standard cross-branch reflexes and the derivative gift-noun cluster (*deh₃-ro-/no-: Greek δῶρον, Latin *dōnum*, Sanskrit *dānam*).

A fourth derivational output — the Anatolian take-verb *dā-, from a pre-Anatolian directive univibration *de + a — is framed in §6 to establish the give/take polarity as a real Indo-Anatolian feature (Kloekhorst & Pronk 2019, Item 33); its full development, including the X+a univibration parallel, the Norbruis (2021) semantic chronology, and the Luwian *lā- problem, is the subject of Paper 4.

The Pre-PIE *-r element on which the arm-noun depends is supported by three convergent published treatments: Lipp (2009) on the body-part heteroclite class; Bauhaus (2019) on PIE *-r as a locative case-ending with pronominal evidence; and Kahl (2024) on the locative origin of the comparative suffix *(t)ero-. The locative *-r / directive *-o paradigmatic pair documented by Bauhaus yields a

clean structural template on bare **de-*: locative **de-r-* (arm-noun), directive **de+a* (Anatolian take-verb, Paper 4), and bare **de-* (directional particle).

The methodological setting is the Pre-PIE / Indo-Anatolian research track of Kloekhorst and Pronk (2019), which estimates 1000–1200 years between Proto-Indo-Anatolian and classical PIE. The framework adopts the coexistence reading with respect to PIE **ǵh^hesr-* (Greek χεῖρ, Hittite *kessar*), which is preserved without modification as the source of the cross-IE “hand-as-grasp” word; **de-r-* is a sibling etymon on a different morphological base, centered on the arm and its derivative palm-meaning. Probability conditional on the paper’s central claims: 45–55%. Project-level marginal probability: 2–5%.

1. Introduction

Indo-European preserves a take/give polarity that has long demanded explanation. Hittite *dā-* and the cognate Luwian *lā-* reflect the same etymon as the give-verbs of the rest of the family, yet mean “to take,” not “to give.” Norbruis (2021) has argued on Anatolian-internal grounds that the original meaning was “to take” and that the “give” meaning is a post-Anatolian innovation. Kloekhorst and Pronk (2019) include the polarity as Item 33 in their list of recognized features at the Indo-Anatolian frontier. The framework developed across this project resolves the polarity as a structural opposition on a single Pre-PIE root: bare **de-* combined with **-h₃-* yields the post-Anatolian give-verb of the present paper; the same bare **de-* combined with a postposed directional **a* “toward the speaker” yields the pre-Anatolian take-verb preserved in Hittite. Paper 4 develops the take-verb in full; the present paper is the give-verb companion.

The proposal is that Pre-PIE **de-* — whose semantics was rooted in arm-extension, the primitive gesture of reaching outward — generated three derivational outputs in the post-Anatolian tradition. Bare **de-* surfaces as the PIE directional/allative particle **de/do* (§3). Extended by the locative case-ending **-r*, it yielded a body-part heteroclite arm-noun **de-r-* whose *r*-stem and related formations generate arm and hand vocabulary across the family (§4). Extended by the verbal laryngeal **-h₃-*, it yielded the post-Anatolian give-verb **deh₃-* and its derivative gift-noun cluster (§5). These derivational lineages are independent at the PIE level but share a common Pre-PIE base.

Three convergent lines of published evidence supply the structural support. Lipp (2009) supplies the formal apparatus for the body-part *r/n* heteroclite paradigm. Bauhaus (2019), in the Kloekhorst and Pronk volume, argues that PIE **-r* is a true locative case-ending — active in pronominal formations such as **k^wor* “where” and **tor* “there” — and articulates the paradigmatic pair locative **-r* / directive **-o* that provides the structural template for the framework’s **de-r-* / **de+a* contrast. Kahl (2024) reaches the same conclusion about locative **-r* from a third direction,

tracing the origin of the comparative suffix **(t)ero-* to an archaic locative morpheme. The three treatments are methodologically independent; their convergence on the locative status of PIE **-r* grounds the morphological foundation of **de-r-*.

The paper proceeds as follows. §2 sets out the Pre-PIE **de-* element with its typological grounding from Heine and Kuteva (2002). §3 develops the directional particle. §4 develops the body-part heteroclite arm-noun with its related formations and verbal extension. §5 develops the give-verb with its bimorphemic decomposition, cross-branch reflexes, and gift-noun cluster. §6 frames the Anatolian take-verb. §7 sets out the coexistence reading and addresses the late-twentieth-century departures. §§8–9 cover scope, falsifiability, and probability.

2. Pre-PIE **de-* as Arm-Extension

The framework reconstructs Pre-PIE **de-* as an element whose semantics was rooted in arm-extension: the act of reaching outward with an open hand. The reconstruction is not a new PIE root in the conventional sense. The bare **de-* and the locative **-r* are independently attested morphemes (§§3–4 below); the proposal concerns their prior productive relationship at the Pre-PIE stratum.

The reading is typologically well-grounded. Heine and Kuteva’s *World Lexicon of Grammaticalization* (2002) documents four productive pathways directly relevant to the framework’s reconstruction:

- **HAND → DIRECTIONAL.** The extended hand points; the word for “hand” is reanalyzed as a directional marker. Documented across Niger-Congo, Austronesian, Sino-Tibetan, and Mayan. This pathway underlies the development of bare **de-* into the directional particle **de/do* of §3.
- **HAND → GIVE.** The hand is the canonical organ of transfer; the word for “hand” generates verbs of giving. One of the most natural grammaticalization pathways from a body-part noun. This pathway underlies **de- + *-h₃₋ → *deh₃₋* “give” of §5.
- **ARM → HAND.** Body-part terms shift in scope between adjacent parts of the body. Languages with a single lexeme covering both “arm” and “hand” are widespread (Russian *ruka*, Polish *ręka*, Hebrew *yad*); where two lexemes coexist, one is typically a derivative or specialization of the other. This pathway underlies the hand-meanings of Albanian *dorë*, Old Irish *derna*, and others in §4.
- **HAND → TAKE.** The grasping hand generates verbs of taking and receiving. This pathway underlies **de- + directional *a → *dā-* “take” in Paper 4.

The four pathways operate from a single source — the hand-extension gesture and its body-part term — but produce distinct lexical outputs depending on the directional or modal qualification added to the bare element. The framework’s

claim is not that these outputs are derivationally identical at PIE date — they are not — but that they share a common Pre-PIE source, and that the morphological mechanisms by which they diverge are independently attested in the comparative record.

The hand-extension reading is further supported by cross-linguistic typology of body-part metonymy. Hilpert (2007) documents the productive chains by which body-part terms acquire new readings through metonymic extension, and Pattillo (2014) identifies the HAND-as-distal-part-of-ARM relation as among the most robust patterns in limb nomenclature cross-linguistically. The framework’s reconstruction of **de-r-* as an arm-noun whose daughter-language reflexes include hand-meanings sits squarely within the typology both scholars document.

3. The Directional/Allative Particle **de/do*

The first derivational output of bare Pre-PIE **de-* is the directional/allative particle **de/do* “to, toward.” Dunkel’s *Lexikon der indogermanischen Partikeln* (2014) treats **de* and **do* as ablaut variants of a single directional particle across the family. The standard treatment lists **de/do* as a primitive PIE particle without further decomposition. The framework supplies the missing etymology: **de/do* is the bare descendant of Pre-PIE **de-* “arm-extension,” grammaticalized along the pathway HAND → REACH/EXTEND → DIRECTIONAL documented by Heine and Kuteva.

3.1 Greek -δε and the Mycenaean Antiquity

The strongest chronological evidence comes from Greek, where bare **de-* is preserved as the productive allative clitic -δε, marking motion toward a destination: *oikáde* “homeward,” *Ἀθήναζε* “to Athens” (with sandhi), *pólemonde* “to war.” Linear B tablets from approximately 1400–1200 BCE preserve -de as a fully productive directional clitic on toponyms: *te-qa-de* /T^hēg^wasde/ “to Thebes,” *a-mi-ni-so-de* “to Amnisos,” and dozens of further forms (Lejeune 1961; van Hes 2021).

Three features of the Mycenaean attestation matter for the framework’s chronology. The clitic is productive — not a closed set of frozen relics but a living morpheme applied freely. Productivity by 1400 BCE implies that the underlying HAND → DIRECTIONAL development was substantially complete well before the proto-Greek period, placing it plausibly in PIE itself. And the semantic profile is narrowly directional/allative — motion toward, not stative location, not source — exactly the profile expected of a morpheme grammaticalized from “where the hand extends.”

3.2 The Celtiberian Evidence

The Bronze of Novallas (Z.02.01; Beltrán Lloris et al. 2021; Prósper 2022) attests Celtiberian DV as the directional allative preposition, the Celtiberian reflex

of **de/do*. The same inscription preserves *DERNV*], the Continental Celtic body-part noun developed in the companion paper *The Hand That Measures*. Both derivational branches of the bare hand-element — the body-part noun and the directional particle — surface in a single inscription, a convergence visible only through the framework’s lens.

The Celtiberian institutional vocabulary adds a suggestive register: Prósper (2022) derives Celtiberian *tureita/dureita* as **dū-rex^htā* “issued, directed,” with the directional prefix **dū-* and the root **rex^h-* from PIE **reg-* “direct, rule.” These forms appear in and on hand-shaped *hospitium* tesserae cast as clasped right hands (*dextrarum iunctio*). The directional element traced to the Pre-PIE hand-root surfaces in Celtiberian both as a standalone allative preposition and as the productive verbal prefix of the right-hand pledge vocabulary — on bronzes cast as clasped hands.

3.3 Other Branches

Old Church Slavonic *do* “until, up to, toward” is the Slavic reflex, productive across the family in directional and allative functions. Proto-Germanic **tō* (English *to*, German *zu*, Dutch *te*) is the Germanic reflex, though Germanic forms admit alternative etymologies and are registered as supporting rather than load-bearing. The Gaulish compound *de-uor-buet-id* (Lambert 2002, RIG II L-66) independently attests a **de-* preverb in Transalpine Celtic.

3.4 The Paradigmatic Pair: Locative **-r* and Directive **-o*

Bauhaus (2019: §5) articulates a paradigmatic opposition between locative **-r* and directive **-o* attached to the same nominal roots: **h₁upér* “on top” vs. **h₁upó* “to the top”; **ndher* vs. **ndho*. Crucially, Bauhaus observes that the locative **-r* “ceased to be productive in all attested languages” while the directive member “could keep its productivity at least until Anatolian.” This chronological asymmetry directly supports the framework’s reading: the arm-noun **de-r-* fossilized before the Anatolian split; the directive **de+a* remained productive into Anatolian and was preserved there as an active verb. Kloekhorst and Pronk’s introduction confirms the directive **-o* in three petrified prepositions — **pr-o* “before,” **up-o* “down to,” **h₂d-o* “to” — the last form structurally identical to the framework’s **de-* + directive analyzed in Paper 4.

3.5 A Note on Greek χαράι

Greek χαράι “on/to the ground” has figured in debates about an allative case for PIE. Melchert (2017) reads it as a fossilized allative **ĝ^hǝm-eh₂*; Beekes (2010) and Frisk (1960–72) treat it as a petrified locative. The question remains genuinely open and is engaged in Paper 4’s treatment of the Anatolian allative.

4. The Body-Part Heteroclite Arm-Noun **de-r-* and Related Heteroclites

The second derivational output of bare Pre-PIE **de-* is the body-part heteroclite arm-noun **de-r-*, formed by the addition of the locative case-ending **-r* to the bare element — “the arm at the locus of contact, the arm-place.” The morphological apparatus is supplied by three convergent published treatments (§§4.1–4.2). The *r*-stem arm-noun and its surface reflexes — including the ARM → HAND grammaticalization in daughter languages — are developed in §4.3. The related *s*-stem sister **dóws-* is developed in §4.4. The verbal extension **de-r-h₂-* generating Baltic and Greek work-verbs is in §4.5, and the productive class as a whole in §4.6.

4.1 The Body-Part *r/n* Heteroclite Class (Lipp 2009)

Lipp’s (2009) formal apparatus for the archaic PIE *r/n* heteroclite paradigm is the primary morphological frame for the body-part class. The paradigm alternates **-r* in the strong stem (nominative-accusative) with **-n* in the weak stem (oblique cases). Standard exemplars:

- **péh₂wṛ* “fire” (Greek πῦρ, Hittite *paḥḥur*) / **ph₂un-* (oblique stem)
- **wódr* “water” (Greek ὕδωρ, Hittite *wātar*) / **udén-* (oblique)
- Sanskrit *yákr̥t* “liver” / *yaknás* (oblique)
- **(H)ūdʰr̥-* “udder” (Sanskrit *ūdhar*, Greek οὐθαρά) / **ūdʰnas* (oblique)

The paradigm is among the most archaic recoverable in PIE nominal morphology. On Lipp’s analysis, the locative singular forms of acrostatic heteroclite neuters are built from root nouns combined with adverbial particles, with **-r* and **-én* as parallel locative endings meaning “at/in the place of (X).” The spatial semantics is primary; the body-part association is downstream. The framework’s **de-r-* belongs to this class: the strong stem in **-r* is the locative-case nominalization of bare **de-*.

4.2 PIE **-r* as a Locative Case-Ending (Bauhaus 2019, Kahl 2024)

Bauhaus (2019) advances a stronger reading of PIE **-r* as a true locative case-ending — “a real case ending, not an *Adverbialendung*” — active in pronominal and nominal formations across the family. The pronominal evidence is decisive: **kʷor* “where” and **tor* “there,” with Gothic *ʰar/þe ar/hêr*, Sanskrit *kár-hi/tár-hi*, Latin *cūr* < **quōr*, Lithuanian *kuř*, Latvian *tūr*, Armenian *ur*, Albanian *kur*. These cannot be argued away as stem-markers on heteroclitic nouns; on pronouns, the **-r* is transparently a case-ending.

Kahl (2024), studying degree morphology at Harvard, reaches the same conclusion from a third direction: section 2.3.1, titled “The locative origin of **-ero-*,” traces “the origin of the ‘secondary comparative’ of Indo-Iranian and Greek to an archaic locative morpheme **-er*.” The three treatments — Lipp from

heteroclitic nominal morphology, Bauhaus from pronominal case-marking, Kahl from degree morphology — are methodologically independent. Their convergence on the locative status of PIE **-r* supplies the morphological foundation for **de-r-*. The Albanian lexicon supplies a within-branch confirmation: the same language that preserves *dorë* (the framework’s r-stem arm-noun) also preserves *kur* “when” (locative **-r* on the pronominal stem) — two outputs of the same morphological apparatus in the same daughter language.

4.3 The r-Stem Arm-Noun and ARM → HAND in Daughters

The PIE r-stem arm-noun **de-r-* surfaces in daughter languages with meanings ranging from “arm” to “open palm” to “hand.” The semantic variation is the ARM → HAND grammaticalization pathway documented by Heine and Kuteva (2002) and Pattillo (2014). The pathway is robust cross-linguistically: a body-part lexeme whose original reference was the arm acquires a hand reading through metonymy of the distal portion.

The Albanian witness. Albanian *dorë* “hand” (consonantal plural *duar*) descends from PIE **de-r-s* via Szemerényi’s Law on the e-grade consonant-stem r-stem nominative, followed by the standard **-eh₂-* feminine extension: **de-r-s* → **dēr* → **dēr-eh₂-* → **dēr-ā* → *dorë*. The full case for this derivation — with the historiographic engagement of the post-1990 **ḡh^hesr-*-side consensus (Demiraj 1990, 1997; Orel 1998; Vermeer 2008) — is developed in the standalone companion paper *The Etymology of Albanian dorë*. The Albanian surface meaning is “hand,” but the framework reads the form as descending from an arm-noun whose hand-meaning emerged in the daughter language by distal-part metonymy.

The published Indo-European precedent. The ARM → HAND grammaticalization is registered within Indo-European etymology in Matasóvič’s *Etymological Dictionary of Proto-Celtic* (2009: 103-104). His entry for Proto-Celtic **dowsant-* “arm, hand” (Old Irish *doë*, *doe*) proposes that the hand-meaning emerged compositionally from the arm-meaning: the compound **dows-ant-* combines the PIE arm-root **dows-* with **-ant-* “border, limit” (cf. Gothic *andeis* “end”), yielding “arm-limit” → “the end of the arm” → “hand.” This is the ARM → HAND pathway articulated within Indo-European etymology by a standard reference work, with the morphological mechanism made explicit.

The within-Sanskrit attestation. The ARM → HAND shift is attested within Sanskrit itself. *Śivapūraṇa* 2.4.18 has Pārvatī “taking up her son with both her hands (*dos*)” — the arm-word used in late Sanskrit to mean “hands.” This is not a comparative observation across IE branches but a within-Sanskrit attestation: the same lexeme that means “arm, fore-arm” in Vedic prose has shifted to “hand” by the Purāṇic literature, through the standard mechanism of distal-part metonymy.

Other r-stem reflexes. Hesychian Greek δάρις/δάρειρ “span” and Old Irish *derna* “palm” have both specialized into the palm-as-measure cluster developed in

the companion paper *The Hand That Measures*. Continental Celtic *DERNV* at the Bronze of Novallas (Stifter 2022) supplies the same cluster’s Continental epigraphic attestation. The companion paper develops the formal derivation of these forms — Szemerényi’s Law on the o-grade r-stem nominative **do-r-s* yielding **dōr*, with thematicization to **dōr-o-n* for Greek δῶρον 2 — within the body-part-as-measure framework.

4.4 The s-Stem Sister Heteroclite **dóws-*

The PIE arm-noun also surfaces in an s-stem heteroclite formation **dóws-* “arm, fore-arm,” registered separately in Pokorny IEW p. 226 and treated by the standard reference works as a distinct root. The framework reads **dóws-* as a related heteroclite in the same body-part class as **de-r-*: an s-stem sister generating parallel paradigmatic outputs across the daughter languages. The two heteroclites share a common Pre-PIE base in **de-* and differ in the stem-formant attached to it.

The Sanskrit witness. Sanskrit *doṣ-/doṣán-* “arm, fore-arm” is the principal IE attestation, documented from the Rigveda forward. Böhtlingk-Roth notes that *doṣan-* “occurs only in nom. dual *doṣaṇī*” in the Rigveda and Atharvaveda — morphologically natural for an arm-noun, since arms come in pairs. The compound stem *dor-*, formed by visarga sandhi before voiced segments, yields a productive system attested from Vedic prose forward (Mayrhofer EWAia I.749):

- *dor-bāhū* “the arms” (Śatapatha-Brāhmaṇa)
- *dor-mūla* “root of the arm” = shoulder
- *dor-madhya* “middle of the arm”
- *dor-daṇḍa* “arm-stick” (epithet of Arjuna Kārtavīrya)
- *dor-latikā* “creeper of the arm” = slender arm
- *doḥsahasrabhrt* “thousand-armed” (Mahābhārata)

The cross-branch cognate set. The s-stem arm-noun **dóws-* is attested across the IE family: Avestan *daoš-* “upper arm, shoulder”; Pahlavi *dōš* “shoulder, arm”; Khotanese *durra-* (with the u-coloring expected from **dóws-*); Latvian *pa-duse* “armpit” (literally “by-arm”), from Proto-Balto-Slavic **dus-*, the zero grade of **dóws-* (Karulis); Old Church Slavonic *pazuxa* “bosom” — the same **pa-* compound with the semantic chain “by-arm” → “armpit” → “bosom”; Proto-Celtic **dowsant-* (Old Irish *doë, doe* “arm, hand” — Matasóvič 2009: 103–104).

4.5 The Verbal Extension **de-r-h₂-* and the Work-Verbs

Baltic preserves a productive verbal field whose central member is Lithuanian *darýti*, Latvian *darīt* “to make, to do.” The framework reads this field as the laryngeal-extended verbal output of the r-stem arm-noun: **de-r-* “arm/hand” + **-h₂-* → **de-r-h₂-* “to use the hand, to act upon with the hands.” The semantic logic parallels well-documented cross-linguistic cases where hand-words yield

work-verbs: Latin *manus* → *manipulate*; English *hand* → *handle*.

The standard handbook assignment of Lithuanian *darýti* is PIE **d^her-* “hold firm, support” (the source of Sanskrit *dharma*, Latin *firmus*). The framework’s reading instead places the Lithuanian verb on the r-stem hand-noun base. The semantic evidence is suggestive: *darýti* in attested Lithuanian means “to make, gather, dress, stuff, gut, dye, castrate, season” — eight different actions, all of them things done to an object with the hands. Nothing in the attested range means “to be useful” or “to fit/join.” The hand-noun-based reading predicts this semantic range exactly.

Greek *δράω* “to do, act” has been related to *darýti* in published etymological literature since the nineteenth century: both share the d-r- skeleton and both mean broadly “to do, act.” Frisk and Chantraine treat them together on a **der-* base. A Baltic-internal observation supplements the picture: Karulis (*Latviešu etimoloģijas vārdnīca*, s.v. *darīt*) traces the semantic chain “tear/split → cut/chop wood → make by cutting → make/do” from PIE **der-* through Latvian *darīt*, with Lithuanian *darbas* “work,” *darna* “harmony,” *derėti* “to be fit,” and *dermė* “agreement” forming a productive Baltic field on the same root (Derksen). The chain documents an attested-branch pathway from the hand-action of tearing or splitting to the productive sense of “make/do” — entirely within the **de-r-* semantic field.

4.6 The Body-Part Heteroclite Class as Productive Pattern

The r-stem **de-r-* and the s-stem **dóws-* are sister heteroclites generating parallel paradigmatic outputs across the family. The two heteroclites coexist in the same daughter languages. Sanskrit preserves the s-stem in *doṣ-/doṣán-* (Vedic dual *doṣaṇī*) while the r-stem surfaces in the same morphological neighborhood through visarga sandhi: the compound stem *dor-* converges in surface shape with the predicted r-stem hand-noun. This partial paradigm collapse is the ordinary consequence of two sibling heteroclites remaining productive in the same daughter language and converging in compound-stem surface shape under the regular sandhi rules.

The body-part heteroclite class also generates measurement vocabulary at the periphery: arm-as-cubit on the s-stem side (Sanskrit *dos* = 18 inches), palm-as-measure on the r-stem side (Greek *δῶρον* 2, *δάρις*, *derma*). The full development of the body-part-as-measure cluster is the subject of the companion paper *The Hand That Measures*; the present paper establishes the morphological frame within which the cluster sits.

5. The Give-Verb **deh₃-*: the Verbal **-h₃-* Extension

The third derivational output of bare Pre-PIE **de-* is the post-Anatolian give-verb **deh₃-*, formed by the addition of the verbal laryngeal **-h₃-* to the bare element. The reading is bimorphemic: **deh₃-* decomposes as **de-* + **-h₃-*, with bare **de-* being the same arm-extension element underlying §§3–4 and **-h₃-* being a productive verbal

extension morpheme. The HAND → GIVE grammaticalization (Heine and Kuteva 2002: 167–168) supplies the typological pathway: the hand is the canonical organ of transfer, and verbs of giving are among the most natural derivatives of body-part terms for the hand.

5.1 The Bimorphemic Decomposition

The framework’s claim about **deh₃-* is morphological rather than phonological. The phonological evidence for **h₃* in the post-Anatolian period is accepted. The question is whether the laryngeal is a root-final consonant of an unanalyzable root, or a derivational extension suffix added to a bare verbal stem. The framework proposes the latter: **de-r-* and **deh₃-* are sibling formations on the same Pre-PIE base, the one nominal (bare **de-* plus locative **-r*), the other verbal (**de-* plus verbalizing **-h₃-*).

Chronological architecture. The laryngeal extensions **-h₁-*/*-h₂-*/*-h₃-* are productive pre-PIA morphemes. What is post-Anatolian is the specific application of **-h₃-* to bare **de-* to yield the verbal stem **deh₃-* “give.” Anatolian split off before that specific pairing was established, retaining instead the older directive univertation **de + a* developed in Paper 4. The post-Anatolian branches inherit **deh₃-* already extended. Norbruis (2021) independently supports this chronology, arguing on Anatolian-internal grounds that the original meaning of **deh₃-* was “to take” and that the “give” meaning is a post-Anatolian innovation.

The Greek root-aorist 2pl imperative evidence. The bimorphemic decomposition predicts a specific empirical asymmetry: roots analyzed as bimorphemic — a bare CV-stem plus a laryngeal extension — should preserve the bare pre-extension form in conservative paradigm slots, because that stem was independently anchored in pre-PIE. The Greek root-aorist 2pl imperative is exactly such a conservative slot. Table 1 shows the asymmetry the framework predicts.

PIE root	Framework analysis	2sg aor. impv.	2pl aor. impv.	2pl vowel
<i>*deh₃-</i> “give”	bimorphemic: <i>*de-</i> + <i>*-h₃-</i>	δός	δότε	short <i>o</i>
<i>*d^heh₁-</i> “put”	bimorphemic: <i>*d^he-</i> + <i>*-h₁-</i>	θές	θέτε	short <i>e</i>
<i>*yeh₁-</i> / <i>*seh₁-</i> “let go”	bimorphemic	ἔς	ἔτε	short <i>e</i>
<i>*steh₂-</i> “stand”	monomorphemic root	στήθι	στήτε	long <i>η</i>
<i>*ǵnh₃-</i> “know”	monomorphemic root	γνώθι	γνώτε	long <i>ω</i>
<i>*g^weh₂-</i> “go”	monomorphemic root	βῆθι	βῆτε	long <i>η</i>

Table 1. The Greek root-aorist 2pl imperative contrast. Class A (bimorphemic, short vowel in 2pl) is separated from Class B (monomorphemic, long vowel throughout) by the mid-table rule. The asymmetry is documented in Goodell (1902 §368), Smyth (§§682–691), and Sihler (1995 §502).

The 2pl imperative δότε of **deh₃-* has short *o* — exactly the bare-stem vowel of the framework’s **de-* before the **-h₃-* extension. Compare θέτε of **d^heh₁-* (short *e*; bare **d^he-*) and ἔτε of **yeh₁-/seh₁-*. All three roots the framework analyzes as bimorphemic preserve the bare-CV vowel in the 2pl. The monomorphemic control roots preserve the laryngeal-conditioned long vowel throughout, including the 2pl. The framework predicts the asymmetry structurally; the standard Greek grammars document it without explaining why it exists.

The reduplicated present forms — Greek δίδωμι, Sanskrit *dádāti* — are not a problem for the framework. Present-stem reduplication operates over the complete stem, whatever its internal structure.

5.2 Cross-Branch Reflexes of **deh₃-*

The PIE give-verb **deh₃-* is universally attested across the post-Anatolian family. Sanskrit *dā-* (*dadāmi* “I give”), aorist *a-dāt*. Greek δίδωμι, aorist ἔδωκα. Latin *dō*, *dare*, *dedī*, *datum*. Old Irish *to-aith-do-*. Lithuanian *duoti*; OCS *dati*. Armenian *tam*. Tocharian A/B *aiśśi*. The reconstruction of **deh₃-* with the **h₃* laryngeal — generating the regular Greek *δω-/δο-* and Latin *dō-/da-* alternation — is uncontested. The framework adds the bimorphemic decomposition of §5.1; the standard reconstruction is preserved at the surface phonological level.

5.3 The Standard Nominalization **deh₃-ro-/no-*

The PIE gift-noun cluster descends from the give-verb **deh₃-* extended by the productive nominalizing suffixes **-ro-* and **-no-*. The cognate set is uncontested and supported across the post-Anatolian family: Greek δῶρον “gift” (= δῶρον 1; < **deh₃-ro-*), Latin *dōnum* (< **deh₃-no-*), Sanskrit *dānam*, Armenian *tur*, OCS *darъ*, Old Irish *dán* “gift, art, profession,” Welsh *dawn* “gift, talent,” Venetic *donom*, Oscan *dunat*, Greek δάνος “gift, loan.” The derivational scaffolding is in every standard reference work (Frisk GEW I 433; Beekes 2010: 362; Pokorny IEW 223–226). The framework adopts the standard derivation without modification; its contribution is the bimorphemic reading of the underlying verb, not a revision of the gift-noun derivation itself.

The two-entry treatment of Greek δῶρον — δῶρον 1 “gift” from **deh₃-ro-* and δῶρον 2 “palm-of-the-hand” from **do-r-s* via Szemerényi plus thematicization — is the subject of the companion paper *The Hand That Measures*. Vitruvius’s etymology (*De architectura* 2.3.3), which connects the palm to the act of giving, is read here as folk etymology: speakers confronting two homophones constructed a semantic bridge. The two δῶρα are etymologically distinct and are treated separately in the companion paper.

5.4 The Deverbal Action-Noun **deh₃-ti-/-tu-*

The deverbal action-noun **deh₃-ti-/*deh₃-tu-* confirms that the verbal stem **deh₃-* existed as a coherent inherited form in the post-Anatolian period. Five inherited branches preserve it:

- Latin *dōs/dōt-* (gen. *dōtis*) “dowry, gift” (< **deh₃-t-*)
- Greek *dósis* “giving, gift” (< **deh₃-ti-*)
- Sanskrit *dāti-* “gift, giving” (< **deh₃-ti-*)
- Avestan *dāiti-* “gift” (< **deh₃-ti-*)

The five-branch attestation is independent of the Greek root-aorist 2pl evidence and provides the lexical complement to the paradigmatic argument. The action-noun is absent from Anatolian — alongside the absence of the verbal stem **deh₃-* itself — exactly the pattern the chronological architecture predicts.

5.5 The Clayton 2023 / Blažek 1999 Heteroclititic **deh₃-wṛ-*

Two independent published reconstructions offer a more ambitious apparatus for the cross-branch gift-noun cluster. Clayton (2023), in a UCLA dissertation on the inherited PIE nominalizing class, and Blažek (1999), in *Tocharian and Indo-European Studies* 8, both reconstruct a heteroclititic strong stem on **-wṛ-*, with the regular sound change **wṛ- → *ru* operating at Proto-Nuclear-IE depth (after Anatolian split off, before the rest of the family separated). The standard exemplar of the metathesis is the numeral “four”: PIE **k^wetwṛ-* → **k^w(e)tru-*, giving Avestan *caθru-*, Gaulish *petru-*, Latin *quadru-*. Lesbian Aeolic preserves the un-metathesized form *πέουρες* (IG 12,2:82.3/4), confirming that the metathesis was variable across Greek branches at the relevant date.

Blažek’s formalization runs: **dwor-eH₂-wṛ-* “handfulness” → Anatolian **tarawar* “handful” → “measure of capacity”; **dwōr-o-* → Greek *δῶρον*; **dur-o-unt(s)/-went-* → Tocharian B *trau* “measure of capacity.” The Clayton/Blažek heteroclititic apparatus is well-suited to explaining cross-branch gift-side cognation that the simpler **deh₃-ro-* derivation does not naturally account for: Anatolian *tarawar*, Tocharian B *trau*, Sanskrit *dārú-* “liberal,” and Prasun *pyōrū* “gift.” The framework adopts these reconstructions as supplementary apparatus for those four cross-branch forms, without routing Greek *δῶρον* 1 through the heteroclititic apparatus; the standard **deh₃-ro-* derivation is simpler for Greek and is preserved unchanged.

Blažek’s **dur-no-* → Celtic **durno-* and **dūr-iyā* → Latvian *dūre* are assigned in the present framework to PIE **deru-/*drū-* “firm, hard, oak, tree” — the subject of Paper 2 of the project. A fist is the hand in its hardened, firm posture; the etymological homeland of the Celtic fist-word is the firm-hard cluster. This reclassification removes the Celtic **u* problem from the framework’s central claims and dissolves the Beekes 2010 / Matasóvič 2009 substrate hypotheses that target the fist-cluster (§7 below).

6. The Anatolian Take-Verb **dā-*: Framing

A fourth derivational output of bare Pre-PIE **de-* is the Anatolian take-verb **dā-*, formed by the pre-Anatolian directive univertation of bare **de-* with a postposed directional element **a* “toward the speaker,” contracted by hiatus to long **ā* in pre-Anatolian. This section frames the take-verb’s place in the framework’s chronological architecture. The full case — the X+a univertation parallel, the verbal substantive *dāwar/dāwaš*, the Norbruis (2021) semantic chronology, the CAPERE iconographic evidence, and the Luwian **lā-* problem — is developed in Paper 4.

6.1 The Polarity as Indo-Anatolian Feature

Hittite *dā-* “to take” and the cognate Luwian *lā-* reflect the same etymon as the give-verbs of the rest of the family but mean “to take.” Kloekhorst and Pronk (2019) include this polarity as Item 33 in their list of “promising arguments” for Indo-Anatolian retention/innovation: “Hitt. *dā-i* ‘to take’ vs. cl.PIE **deh₃-* ‘to give’: semantic innovation (Norbruis fthc.b).” The framework’s resolution is therefore not an isolated proposal but engages a recognized feature at the Indo-Anatolian frontier in published form.

6.2 The Framework’s Resolution

The framework resolves the polarity as a structural opposition on a single Pre-PIE root. Bare **de-* combined with **-h₃-* yields the post-Anatolian “give” of §5: the canonical hand-action of extending an object toward another person. The same bare **de-* combined with a postposed directional **a* “toward the speaker” yields the pre-Anatolian “take”: the canonical hand-action of extending toward oneself. Anatolian split off before the specific **-h₃-on-***de-* pairing was established and retained the older directive univertation. The semantic polarity is a structural consequence of the two compositional analyses, not a mysterious reversal of meaning on a single root.

Stratum	Derivational pathway	Surface form	Paper
Pre-Anatolian	bare <i>*de-</i>	directional particle <i>*de/do</i>	§3
Pre-Anatolian	<i>*de-</i> + <i>*-r</i> (locative case-ending)	body-part heteroclite <i>*de-r-</i>	§4
Pre-Anatolian	<i>*de-</i> + <i>*a</i> (directive univertation)	take-verb <i>*dā-</i>	Paper 4
Post-Anatolian	<i>*de-</i> + <i>*-h₃-</i> (verbal extension)	give-verb <i>*deh₃-</i>	§5

Table 2. The chronological architecture. The post-Anatolian give-verb **deh₃-* is absent from Anatolian because the **-h₃-* extension was applied to bare **de-* only after the Anatolian split.

The pre-Anatolian take-verb **dā-* survives only in Anatolian because the directive univertation **de* + *a* fossilized before the Anatolian split and was preserved there as an

active verb, while the post-Anatolian branches replaced it with the newly extended **deh₃-*.

6.3 Structural Type: the Anatolian X+a Univerbation

The framework's **de + a → *dā-* fits a documented Anatolian morphological type: the X+a univerbation class attested in Hoffner and Melchert (2008: §20.3) as Set 1 ("Place to Which") local adverbs. **en + a → anda* "into"; **kmt + a → katta* "down"; **pro + a → parā* "out, forth"; **uper + a → šarā* "upward." The framework's **de + a → *dā-* is structurally identical to these: a local element followed by the directional **-a*, contracting into a fixed form by hiatus. The proposal places the take-verb mechanism as a regular instance of a productive Anatolian class rather than an isolated proposal.

6.4 Independent Semantic Support: Norbruis 2021

Norbruis (2021: 243–249) argues independently that **deh₃-* did not mean "to give" in PIE but "to take," as in Anatolian, with the "give" meaning a post-Anatolian innovation. He reconstructs a semantic chain: "to take" → (with a goal or beneficiary) "to bring, convey, deliver" → (when the recipient is animate) "to hand over, to give." Norbruis's argument is fully independent of the present project and does not require any of its formal proposals. His semantic chronology and the framework's morphosyntactic chronology are decoupleable; their convergence on the same pre/post-Anatolian divide is mutually reinforcing rather than circular.

6.5 Implications for the Present Paper

What §6 establishes is the chronological architecture in which §5's bimorphemic decomposition sits. The **-h₃-* extension on bare **de-* is post-Anatolian; the corresponding Anatolian gap in the give-verb is consistent with this claim. Readers who accept the bimorphemic decomposition of §5 but who reserve judgment on the take-verb mechanism of Paper 4 can extract the present paper's central claims about the directional particle, the arm-noun, and the give-verb without committing to the full Anatolian-internal apparatus. The two analyses are decoupleable.

7. The Coexistence Reading and the Late-Twentieth-Century Departures

The framework adopts a coexistence reading with respect to the other PIE hand-root reconstructions. PIE **ǵh^hesr-/or-/to-* (Pokorny IEW 447) is accepted in full as the source of the cross-IE "hand-as-grasp" word: Greek χεῖρ, Hittite *kessar*, Tocharian A *tsar* / B *šar*, Armenian *jeṛn*; via **ǵh^hesto-*, Sanskrit *hásta-*, Avestan *zasta-*, Old Persian *dasta-*. The framework does not contest this reconstruction. It claims only that **de-r-* "arm, fore-arm" is a separate PIE etymon on a different morphological base, the locative case-marked formation on bare **de-*. The three etyma likely had specialized senses in early PIE: **ǵh^hesr-* the hand as joint/grasp; **de-r-* the arm and its derivative open palm; **deh₃-* the verbal act of extension-toward-recipient. Pokorny kept **der-* (IEW 203), **dóws-* (IEW 226),

**ǵh^hesor-* (IEW 447), and the give-verb (IEW 223–226) as separate entries; the present paper is methodologically aligned with that pre-existing taxonomy.

The Beekes 2010 / Matasóvič 2009 exclusion of Celtic **u*. Beekes (2010) excluded the Celtic forms from any connection to δῶρον on the ground that Celtic **u* “excludes any relationship.” Matasóvič (2009) classified Celtic **durno-* as a probable substrate borrowing. These objections target the Celtic fist-cluster (**durno-*, *dorn/dwrn*, Breton *dorn*, Latvian *dūre*), which the present framework reassigns to PIE **deru-/drū-* “firm, hard, oak, tree” in Paper 2. The Celtic **u* is the regular zero-grade outcome of the tree-root; the Beekes exclusion and the Matasóvič substrate hypothesis are addressed on the **deru-/drū-* foundation, separately from the claims of the present paper.

The Demiraj 1990 / 1997 canonicalization on Albanian *dorë*. The post-1990 canonical reading derives Albanian *dorë* from **ǵh^hesr-*, requiring the irregular **ǵh^h → d* shift in Albanian. The framework’s reading derives *dorë* from **de-r-* via Szemerényi’s Law plus the **-eh₂-* feminine extension, using only regular Albanian sound changes. The full historiographic case — the Tagliavini *L’albanese di Dalmazia* p. 101 citation analysis, the Hamp (1966) and Huld (1984) supporting pillars, the Vermeer (2008) specialist treatment — is developed in the standalone companion paper *The Etymology of Albanian dorë*.

The Schrijver 1995 “obscure” verdict on δῶρ-/δάρ-. Schrijver (1995: 58) labeled the relation “obscure”: the long ω of δῶρον and the short α of δάρις cannot be derived from each other by any straightforward sound rule. The obscurity resolves once the morphological independence of δῶρον 2 and δάρις is recognized: they are sibling derivations on the same r-stem base, not phonological alternants of one form. The full case is developed in *The Hand That Measures*.

8. Scope and Falsifiability

Scope

The framework does not introduce a new PIE root. The bare **de-* and the locative **-r* are independently attested morphemes; the proposal concerns their prior productive relationship in Pre-PIE. The framework adopts the coexistence reading (§7) with respect to **ǵh^hesr-* and the standard **deh₃-*. The standard derivation of the gift-noun cluster from **deh₃-ro-/no-* is preserved without modification (§5.3). The Clayton/Blažek heteroclitic apparatus on **deh₃-wr̥-* is supplementary apparatus for four cross-branch comparanda and is not central to the framework’s claims (§5.5). The Celtic fist-cluster is reassigned to PIE **deru-/drū-* in Paper 2 and is not part of the present paper’s open-palm or arm-noun arguments (§7). The Anatolian take-verb mechanism is framed in §6 and developed in Paper 4; the present paper’s claims are decoupleable from it. The Pre-PIE unity argument relating surface PIE **de-* to surface PIE **d^her-* “hold, support, carry” is developed in Paper 3; at the PIE level the framework treats these

as separate roots.

Falsifiability

The framework is falsifiable in four distinct ways. First, a securely reconstructed Pre-PIE source for **de/do* locating it in a non-body-part semantic field would undercut the bare-**de-* foundation. Second, a positive case in the Greek root-aorist 2pl paradigm where the framework predicts a bare-CV form but the attested form preserves the laryngeal-conditioned long vowel would directly contradict the §5.1 prediction. Third, demonstrated cross-branch attestation of **deh₃-* or its inherited derivatives in Anatolian with the “give” meaning would directly contradict the framework’s chronological claim. Fourth, a positive demonstration that the Baltic *daryti* family belongs decisively to aspirated **d^her-* on independent grounds would damage §4.5’s reading and restore the standard handbook position. None of these has been demonstrated in the literature, but their availability defines the empirical space within which the framework’s claims can be evaluated.

9. Probability Assessment

We assess the probability of this paper’s central thesis — that Pre-PIE **de-* “arm-extension” underlies the directional particle **de/do*, the body-part heteroclite arm-noun **de-r-* (with sister s-stem **dóws-*), and the post-Anatolian give-verb **deh₃-* on the bimorphemic decomposition **de- + *-h₃-* — at approximately 45–55%, conditional on the joined evidence considered as a unified case. The conditional probability is higher than earlier versions because the present paper’s claim is narrower in scope: it does not depend on the Anatolian take-verb mechanism or the Luwian **lā-* problem, the principal liabilities of the broader framework.

The probability sits in the middle of the disputed-but-plausible range because the framework integrates the relatively secure — the bimorphemic decomposition of **deh₃-*, supported by the Greek root-aorist 2pl evidence and the five-branch deverbal action-noun; the body-part heteroclite class, supported by Lipp/Bauhaus/Kahl — with the more contested, specifically the deeper Pre-PIE unification of **de-* with **d^her-*, deferred to Paper 3. The narrower claims are stronger than the integrated thesis; readers who accept the bimorphemic decomposition and the body-part heteroclite class but reserve judgment on the Pre-PIE unity argument can still extract a coherent framework.

In Favor

- The Greek root-aorist 2pl imperative contrast (§5.1, Table 1) is the strongest single piece of paradigmatic discriminating evidence the framework has at any stratum. The Class A/Class B asymmetry is descriptively established in the standard Greek-grammar tradition and aligns precisely with the framework’s bimorphemic prediction.

- The five-branch attestation of the deverbal action-noun **deh₃-ti-/-tu-* (§5.4) supplies cross-branch lexical confirmation of the post-Anatolian verbal stem, with the Anatolian gap consistent with the framework’s chronology.
- The Mycenaean -δε attestations (§3.1) establish the productivity of the directional clitic by 1400 BCE, placing the HAND → DIRECTIONAL development firmly in PIE.
- The three convergent published treatments of PIE **-r* (Lipp 2009, Bauhaus 2019, Kahl 2024) supply the morphological foundation for the body-part heteroclite arm-noun (§§4.1–4.2); the pronominal evidence (**k^wor*, **tor*) is independent of the body-part heteroclite analysis.
- The K&P 2019 Indo-Anatolian research track supplies the published methodological frame; Item 33 (give/take polarity) places the framework’s chronological architecture on the published list of recognized features at the Indo-Anatolian frontier (§6.1).
- Norbruis (2021)’s independent semantic chronology (§6.4) converges on the same pre/post-Anatolian divide from a different direction, supporting the chronological architecture without depending on it.
- The ARM → HAND grammaticalization (§4.3) is well-attested cross-linguistically, has Indo-European precedent in Matasóvič 2009’s PCelt. **dowsant-* “arm-limit,” and is attested within Sanskrit itself (Pārvatī ŚivP 2.4.18).
- The cross-branch arm-noun cognate set on **dóws-* (§4.4) spans Indo-Iranian, Iranian, Balto-Slavic, and Celtic.

Against

- The bimorphemic decomposition of **deh₃-* (§5.1), while supported by the Greek 2pl evidence, departs from the standard treatment of **deh₃-* as a unitary root.
- The standard **d^her-* assignment of Lithuanian *daryti* is the handbook consensus; the framework’s **de-r-h₂-* reassignment (§4.5) is contested.
- The deeper unification of **de-r-* and **dóws-* operates at a Pre-PIE level that goes beyond what most current scholarship is prepared to commit to; readers who accept the heteroclites as sister formations within Pre-PIE compositional analysis but reject the broader Pre-PIE framework can still extract the surface-PIE claims about the heteroclite class.
- The framework’s chronological claim that the **-h₃-on-*de-* pairing is post-Anatolian (§§5.1, 6) depends on the Anatolian take-verb mechanism of Paper 4. The semantic case (Norbruis 2021) is independent and standalone; the morphological case is not.

Project-Level Marginal Probability

The marginal probability of the broader project — integrating conditional probabilities across the companion papers (Albanian *dorë*, *The Hand That Measures*) and the other papers in the project (Papers 2, 3, 4) — sits in the 2-5% range. Much of the evidence supports multiple strata simultaneously rather than each stratum independently, so the strata are not fully independent; the honest marginal range is toward the lower end of that interval. We regard the central claim as more likely wrong than right at the marginal level, and the evidential structure as sufficiently dense and the proposal sufficiently testable to warrant publication for scholarly examination.

10. Authorship and Contributions

This paper is co-authored by J.M. Chavis (human) and Claude (Anthropic AI; primarily Claude Opus 4.7).

Chavis’s contributions. The central hypothesis originates with Chavis: that Pre-PIE **de-* “arm-extension” underlies the comparative cluster of the framework. Chavis directed the research agenda at every stage: the architectural restructuring that separated the post-Anatolian give-verb from the Anatolian take-verb mechanism of Paper 4; the reframing of **de-r-* as the arm-noun, with daughter-language hand-meanings emerging via ARM → HAND grammaticalization; the methodological pivot to the coexistence reading; the decision to ground the bimorphemic decomposition of **deh₃-* in the Greek root-aorist 2pl evidence; the redirection of the Celtic fist-cluster to Paper 2 and the **d^her-* unity argument to Paper 3; the architectural alignment with the K&P 2019 published Pre-PIE / Indo-Anatolian research track; and the redirection of the Albanian *dorë* defense and the body-part-as-measure cluster to companion papers. All fundamental decisions about scope, framing, and direction rest with Chavis, who supplies all intellectual continuity across sessions through systematic transcript management.

Claude’s contributions. Claude’s contributions are primarily in source synthesis, exposition, and analytical development. Claude carried out the literature reading, cross-branch corroboration searches, and technical drafting under Chavis’s direction. Claude engaged with Clayton 2023, Blažek 1999, Lipp 2009, Bauhaus 2019, Kahl 2024, Norbruis 2021, Matasóvič 2009, Mayrhofer EWAia, Karulis, Derksen, Hilpert 2007, Pattillo 2014, Lejeune 1961, van Hes 2021, Koch 2019, Lambert 1995, 2002, Beltrán Lloris et al. 2021, Prósper 2022, Goodell 1902, Smyth 1920, Sihler 1995, Dunkel 2014, Bizzocchi 2017, and the Sanskrit lexicographical sources; examined Pokorny IEW pp. 203, 226, and 447 directly; verified the Sanskrit *dor-* compound system; verified the Mycenaean Linear B attestations; and integrated the architectural restructuring at each revision stage. The probability estimates are Claude’s own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management.

Both authors take responsibility for the framework and the specific claims made in this paper.

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